Quantification: 543_z13
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	11
interior lacuna count	11
frame-edge lacuna count	0

Interior lacunae only: mean, median and sample SD (n = 11)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	1862	1736	823.012	px^2
major_axis_length_px	lacuna shape	78.3615	74.0218	22.3719	px
minor_axis_length_px	lacuna shape	30.8137	31.6227	6.9101	px
aspect_ratio	lacuna shape	2.5873	2.6086	0.6752	unitless
eccentricity	lacuna shape	0.9006	0.9236	0.0676	unitless
solidity	lacuna shape	0.9037	0.9231	0.0388	unitless
roots_count	headline	5.7273	6	1.954	count
ring_length_r30_px	headline	274.091	281	62.5083	px
ring_length_r60_px	reported	830.182	819	169.533	px
owned_length_px	ownership-dependent	2254.91	1814.63	1416.9	px
edge_count	ownership-dependent	95.6364	79	64.8047	count
mean_edge_length_px	ownership-dependent	25.017	23.5144	3.8232	px
perimeter_px	normalised	207.479	205.037	54.0901	px
ring_area_r30_px2	normalised	8447.82	8219	1466.9	px^2
ring_area_r60_px2	normalised	21999.7	21794	3275.21	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	1	1	0	unitless
ring_density_r30	normalised	0.0330318	0.0337095	0.00792469	px^-1
ring_density_r60	normalised	0.0381178	0.0408581	0.00747877	px^-1
roots_per_100px_perimeter	normalised	2.8038	2.8301	0.8448	per 100 px
ring_attached_length_r30_px	network v2	205.454	210	63.5537	px
ring_attached_length_r60_px	network v2	516.455	507	220.782	px
ring_length_w_r30_px	network v2	310.752	305.948	70.378	px
ring_length_w_r60_px	network v2	939.548	925.98	193.457	px
sholl_crossings_r10	network v2	6.5455	7	2.3394	count
sholl_crossings_r20	network v2	10.0909	10	2.8091	count
sholl_crossings_r30	network v2	12.4545	13	2.4234	count
ring_width_mean_r30_px	width	6.0179	6.1511	0.4217	px

Per field
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measure	value	units
analysed_area_px2	1.02809e+06	px^2
total_skeleton_length_px	36701	px
canalicular_length_density_per_px	0.0356981	px^-1
junction_count	743	count
junction_density_per_px2	0.0007227	px^-2
lacuna_count	11	count
lacunae_per_px2	1.07e-05	px^-2
skeleton_component_count	259	count

mean_component_length_px	141.703	px
median_component_length_px	26	px
field_length_density_w_per_px	0.0405495	px^-1
field_density_without_flagged_per_px	0.0356981	px^-1
field_density_in_roi_per_px		px^-1

Canalicular width, the new measure

measure	value	units
width_mean_px	6.908	px
width_median_px	7.2111	px
width_p10_px	4	px
width_p90_px	8.9443	px
width_pixels_used	36701	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2019	79.1535	33.2822	2.3783	0.9073	0.9417	0.4378	97.9678	270.286	False
2	839	61.6665	17.863	3.4522	0.9571	0.906	0.068	131.349	828.439	False
3	1736	73.3356	33.3815	2.1969	0.8904	0.8374	0.2312	218.82	919.248	False
4	1138	46.8317	31.6227	1.481	0.7376	0.9252	0.396	263.104	340.498	False
5	2267	87.9083	33.2852	2.6411	0.9255	0.9473	0.0042	459.621	481.093	False
6	860	45.4745	26.4762	1.7176	0.813	0.854	0.07	475.221	748.086	False
7	2150	103.918	27.5311	3.7745	0.9643	0.9231	0.2711	546.673	873.734	False
8	3069	109.359	37.084	2.9489	0.9407	0.9099	0.049	642.67	188.33	False
9	1616	74.0218	28.3763	2.6086	0.9236	0.9245	-0.0477	731.536	630.238	False
10	1438	72.5092	25.5367	2.8394	0.9359	0.9236	0.0953	793.137	478.379	False
11	3350	107.799	44.5116	2.4218	0.9108	0.8475	-0.088	854.917	288.104	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	6	256	764	3223.04	145	22.2279	6.3462
2	5	281	693	3092	120	25.7667	5.7936
3	9	310	819	1726.19	71	24.3125	5.655
4	3	227	855	866.607	38	22.8055	6.1511
5	7	326	1012	3937.22	170	23.1601	6.2124
6	4	310	889	2774.7	118	23.5144	5.9959
7	6	293	1005	1177.8	50	23.5561	5.418
8	5	205	705	1814.63	79	22.97	6.3804
9	3	186	590	1091.65	33	33.0803	6.2749
10	7	222	662	286.309	9	31.8121	5.3237
11	8	399	1138	4813.85	219	21.981	6.6452

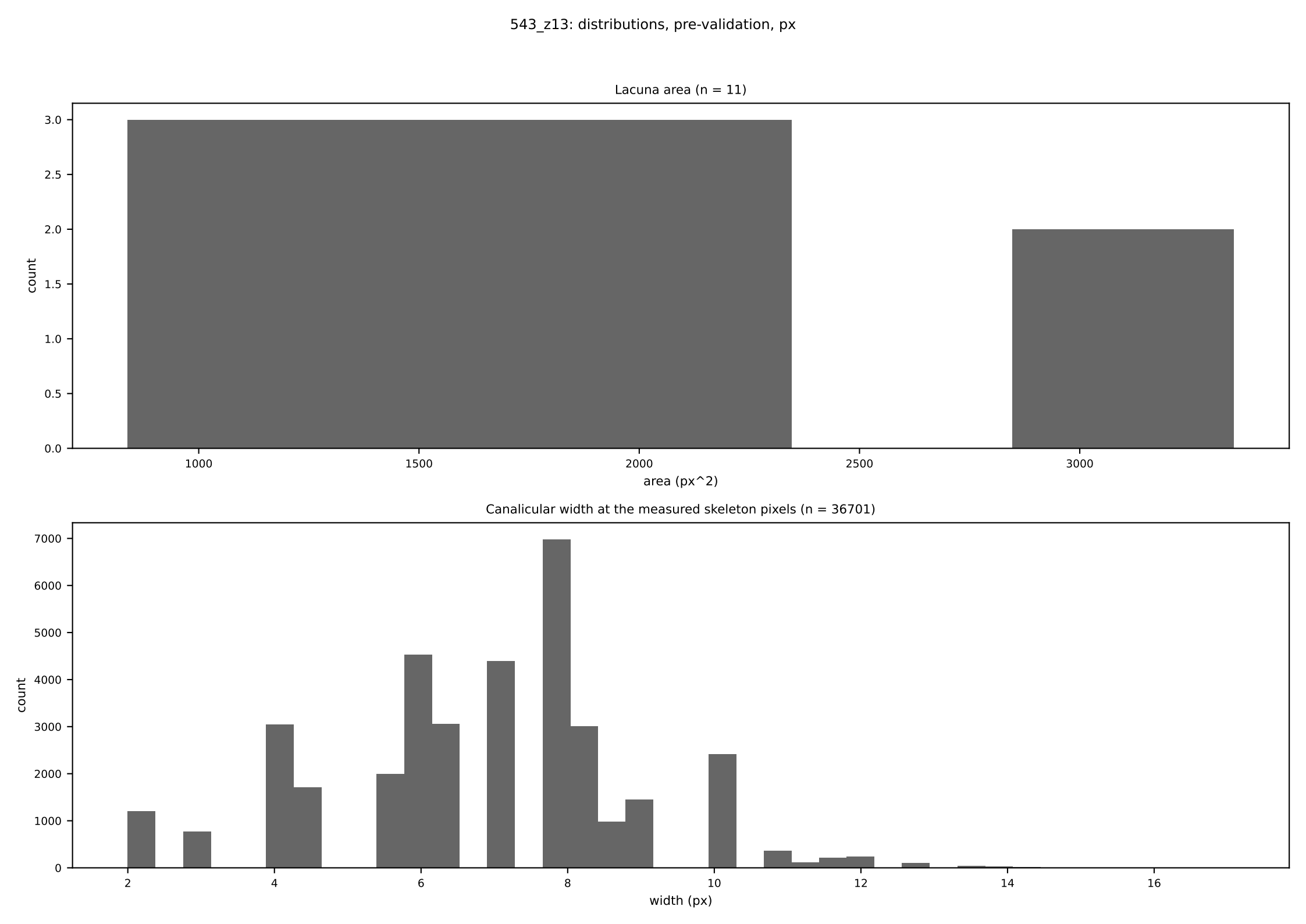
Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	205.037	8499	22530	1	1	0.0301212	0.0339103	2.9263

2	151.297	6982	17707	1	1	0.0402464	0.0391371	3.3048	
3	205.037	8219	20045	1	1	0.0377175	0.0408581	4.3895	
4	139.681	6734	18999	1	1	0.0337095	0.0450024	2.1477	
5	223.823	8900	23453	1	1	0.0366292	0.0431501	3.1275	
6	141.338	6516	17817	1	1	0.0475752	0.0498962	2.8301	
7	254.936	9810	24446	1	1	0.0298675	0.041111	2.3535	
8	278.593	10387	26403	1	1	0.0197362	0.0267015	1.7947	
9	195.723	8083	21794	1	1	0.0230113	0.0270717	1.5328	
10	184.995	7835	21305	1	1	0.0283344	0.0310725	3.7839	
11	301.806	10961	27498	1	1	0.0364018	0.0413848	2.6507	

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	235	520	305.948	881.749	8	10	10
2	238	507	304.924	755.86	6	11	11
3	210	510	351.635	925.98	10	12	13
4	128	320	261.208	964.909	3	8	13
5	219	652	369.049	1139.75	7	12	17
6	268	683	345.279	1004.57	6	11	13
7	188	350	331.563	1138.08	7	10	13
8	162	494	240.865	815.323	7	8	9
9	112	338	207.953	671.5	3	7	11
10	168	263	245.51	742.227	5	6	11
11	332	1044	454.333	1295.09	10	16	16


543_z13: distributions, pre-validation, px
Lacuna area (n = 11)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 36701)
count
width (px)